Step-by-step derivation for an ideal non-interacting Gaussian chain correlation

Model and notation

We consider a discrete Gaussian chain with sites $\mathbf{r}_0, \mathbf{r}_1, \ldots, \mathbf{r}_N$. Define bond vectors $\mathbf{u}_k = \mathbf{r}_k - \mathbf{r}_{k-1}$ (for $k = 1, \ldots, N$). Each bond is independently Gaussian with kuhn length b_k^2 per vector, i.e. the configurational weight is

$$\mathcal{W}[\{\mathbf{r}_k\}] = \exp\left[-\frac{1}{2}\sum_{k=1}^{N}\frac{3}{b_k^2}\big(\mathbf{r}_k - \mathbf{r}_{k-1}\big)^2\right]. \tag{1}$$

The normalization (partition function) is

$$Z_0 = \int \prod_{k=0}^{N} d\mathbf{r}_k \; \mathcal{W}[\{\mathbf{r}_k\}], \tag{2}$$

and we assume a periodic box of volume V.

Quantity of interest

We want to evaluate the correlation of monomers (representing by delta function) i and j at $\mathbf{r}$ and $\mathbf{r}'$:

$$R_{ij}(\mathbf{r}, \mathbf{r}') = \big\langle \delta(\mathbf{r} - \mathbf{r}_i)\, \delta(\mathbf{r}' - \mathbf{r}_j) \big\rangle_0 = \frac{1}{Z_0}\int \prod_{k=0}^{N} d\mathbf{r}_k \; \delta(\mathbf{r} - \mathbf{r}_i)\, \delta(\mathbf{r}' - \mathbf{r}_j)\, \mathcal{W}. \tag{3}$$

Step 1: Fourierify the delta constraints

Use $\delta(\mathbf{x}) = \int \frac{d\mathbf{q}}{(2\pi)^3} e^{i\mathbf{q}\cdot\mathbf{x}}$ for both deltas (we assume the fourier space is also a continuous space):

$$\begin{aligned} R_{ij}(\mathbf{r}, \mathbf{r}') = \frac{1}{Z_0}\int \frac{d\mathbf{q}}{(2\pi)^3}\int \frac{d\mathbf{q}'}{(2\pi)^3}\; e^{\,i\mathbf{q}\cdot\mathbf{r} + i\mathbf{q}'\cdot\mathbf{r}'} \\ \times \int \prod_{k=0}^{N} d\mathbf{r}_k \; \exp\left\{ -\, i\mathbf{q}\cdot\mathbf{r}_i - \, i\mathbf{q}'\cdot\mathbf{r}_j - \frac{1}{2}\sum_{k=1}^{N}\frac{3}{b_k^2}\big(\mathbf{r}_k - \mathbf{r}_{k-1}\big)^2 \right\}. \end{aligned} \tag{4}$$

The advantage is that the configuration integral becomes Gaussian.

Step 2: Change variables to bonds (unit Jacobian)

Let $\mathbf{u}_k = \mathbf{r}_k - \mathbf{r}_{k-1}$ for $k \geq 1$ and keep $\mathbf{r}_0$. Then

$$\mathbf{r}_i = \mathbf{r}_0 + \sum_{k=1}^{i}\mathbf{u}_k, \qquad \mathbf{r}_j = \mathbf{r}_0 + \sum_{k=1}^{j}\mathbf{u}_k. \tag{5}$$

The Jacobian of $\{\mathbf{r}_1,\dots,\mathbf{r}_N\}\mapsto\{\mathbf{u}_1,\dots,\mathbf{u}_N\}$ is 1. The exponent linear in positions becomes

$$-\,i\mathbf{q}\cdot\mathbf{r}_i-\,i\mathbf{q}'\cdot\mathbf{r}_j=-\,i(\mathbf{q}+\mathbf{q}')\cdot\mathbf{r}_0-\,i\mathbf{q}\cdot\sum_{k=1}^{i}\mathbf{u}_k-\,i\mathbf{q}'\cdot\sum_{k=1}^{j}\mathbf{u}_k. \tag{6}$$

The quadratic part depends only on $\{\mathbf{u}_k\}$:

$$\sum_{k=1}^{N}\frac{3}{b_k^2}\big(\mathbf{r}_k-\mathbf{r}_{k-1}\big)^2=\sum_{k=1}^{N}\frac{3}{b_k^2}\,\mathbf{u}_k^2. \tag{7}$$

Step 3: Integrate out the reference position $\mathbf{r}_0$

The integral over $\mathbf{r}_0$ is immediate:

$$\int d\mathbf{r}_0\,\exp\big[-\,i(\mathbf{q}+\mathbf{q}')\cdot\mathbf{r}_0\big]=(2\pi)^3\,\delta(\mathbf{q}+\mathbf{q}'), \tag{8}$$

so the $\mathbf{q}'$-integral collapses and $\mathbf{q}'\to-\mathbf{q}$. We henceforth take $j\ge i$ (the result is symmetric in i,j and depends on $|j-i|$ only).

Step 4: Identify which bonds couple to q

From (6) with $\mathbf{q}'=-\mathbf{q}$,

$$-\,i\mathbf{q}\cdot\sum_{k=1}^{i}\mathbf{u}_k+\,i\mathbf{q}\cdot\sum_{k=1}^{j}\mathbf{u}_k=i\mathbf{q}\cdot\sum_{k=i+1}^{j}\mathbf{u}_k, \tag{9}$$

so only the bonds $k=i+1,\dots,j$ carry a linear source $i\mathbf{q}\cdot\mathbf{u}_k$.

Step 5: Factorize the bond integrals

The integral now factorizes over bonds:

$$\begin{aligned}\mathcal{I}(\mathbf{q})&=\int\prod_{k=1}^{N}d\mathbf{u}_k\,\exp\left\{i\mathbf{q}\cdot\sum_{k=i+1}^{j}\mathbf{u}_k-\frac{1}{2}\sum_{k=1}^{N}\frac{3}{b_k^2}\mathbf{u}_k^2\right\}\\&=\prod_{k\le i\ \text{or}\ k\ge j+1}\underbrace{\int d\mathbf{u}_k\,\exp\left(-\frac{3}{2b_k^2}\mathbf{u}_k^2\right)}_{I_0(b_k)}\times\prod_{k=i+1}^{j}\underbrace{\int d\mathbf{u}_k\,\exp\left(i\mathbf{q}\cdot\mathbf{u}_k-\frac{3}{2b_k^2}\mathbf{u}_k^2\right)}_{I_1(b_k,\mathbf{q})}.\end{aligned} \tag{10}$$

The two elementary Gaussian integrals are

$$I_0(b)=\int d\mathbf{u}\,e^{-\frac{3}{2b^2}\mathbf{u}^2}=\left(\frac{2\pi b^2}{3}\right)^{3/2},\qquad I_1(b,\mathbf{q})=\int d\mathbf{u}\,e^{i\mathbf{q}\cdot\mathbf{u}-\frac{3}{2b^2}\mathbf{u}^2}=\left(\frac{2\pi b^2}{3}\right)^{3/2}e^{-\frac{b^2q^2}{6}}. \tag{11}$$

Step 6: Evaluate Z_0 and cancel the normalization

The partition function in bond variables is

$$Z_0=\int d\mathbf{r}_0\prod_{k=1}^{N}d\mathbf{u}_k\,e^{-\frac{3}{2b_k^2}\mathbf{u}_k^2}=V\prod_{k=1}^{N}\left(\frac{2\pi b_k^2}{3}\right)^{3/2}. \tag{12}$$

Combining (4)–(11) and dividing by Z_0, all prefactors $\left(2\pi b_k^2/3\right)^{3/2}$ cancel exactly, leaving only the $\mathbf{q}$-dependent factor $\prod_{k=i+1}^{j} e^{-b_k^2 q^2/6}$. Define the connector mean-square length

$$\Sigma_{ij} := \sum_{k=i+1}^{j} \frac{b_k^2}{3}, \tag{13}$$

so that the remaining factor is $e^{-\frac{1}{2}\Sigma_{ij} q^2}$.

Step 7: Perform the final q-integral

We have reduced the result to a single Gaussian integral:

$$R_{ij}(\mathbf{r}, \mathbf{r}') = \frac{1}{V} \int \frac{d\mathbf{q}}{(2\pi)^3} \, \exp\left[i\mathbf{q} \cdot (\mathbf{r} - \mathbf{r}') - \frac{1}{2}\Sigma_{ij} q^2 \right]. \tag{14}$$

Using the standard identity

$$\int \frac{d\mathbf{q}}{(2\pi)^3} \, e^{i\mathbf{q}\cdot\mathbf{s} - \frac{1}{2}\Sigma_{ij} q^2} = \frac{1}{(2\pi)^3} \left(\frac{2\pi}{\Sigma_{ij}}\right)^{3/2} \exp\left[-\frac{\mathbf{s}^2}{2\Sigma_{ij}} \right], \tag{15}$$

with $\mathbf{s} = \mathbf{r} - \mathbf{r}'$, we obtain the closed form

$$R_{ij}(\mathbf{r}, \mathbf{r}') = R_{ij}(\mathbf{r} - \mathbf{r}') = \frac{1}{V} \left(\frac{3}{2\pi \sum_{k=i+1}^{j} b_k^2} \right)^{3/2} \exp\left[-\frac{3\,|\mathbf{r} - \mathbf{r}'|^2}{2 \sum_{k=i+1}^{j} b_k^2} \right], \tag{16}$$

Checks and comments.

- **Translational & rotational invariance:** R_{ij} depends only on $|\mathbf{r} - \mathbf{r}'|$ and not on absolute positions or orientation.
- **Symmetry in indices:** For $i > j$ replace $\sum_{k=i+1}^{j}$ by $\sum_{k=j+1}^{i}$; the result is a function of $|i - j|$ only.
- **Bond discussion:** For a homogeneous polymer with identical Kuhn length $b_k = b$ for all k, we have

  $$\sum_{k=i+1}^{j} b_k^2 = (j - i) b^2.$$

  In contrast, for an asymmetric diblock copolymer where the ratio of Kuhn lengths is defined as $\epsilon = b_A/b_B$, the summation becomes

  $$\sum_{k=i+1}^{j} b_k^2 = \sum_{k=i+1}^{fN} \epsilon b_B^2 + \sum_{k=fN+1}^{j} b_B^2 = (fN - i)\, \epsilon b_B^2 + (j - fN)\, b_B^2.$$

  More general cases can be derived by following the same reasoning.

Density–density correlation for μ and ν

We now compute the two–point density correlation for species μ and ν

$$T_{\mu\nu}(\mathbf{r}, \mathbf{r}') = \left\langle \rho_\mu(\mathbf{r})\, \rho_\nu(\mathbf{r}') \right\rangle_0 \tag{17}$$

where ρ_μ is the reduced microscopic density of species μ is defined as:

$$\rho_\mu = \frac{V}{N} \sum_{i=0}^{N_\mu} \delta(\mathbf{r} - \mathbf{r}_i) \tag{18}$$

Writing the densities as sums of delta functions over monomer positions gives

$$T_{\mu\nu}(\mathbf{r}, \mathbf{r}') = \frac{V^2}{N^2} \sum_{i=0}^{N_\mu} \sum_{j=0}^{N_\nu} \Big\langle \delta(\mathbf{r} - \mathbf{r}_i)\, \delta(\mathbf{r}' - \mathbf{r}_j) \Big\rangle_0. \tag{19}$$

Same–species case and use of the derivation R_{ij}

Focusing on correlations within the same species ($\mu = \nu$) and using the symmetry $(i, j) \leftrightarrow (j, i$, we reduce the double sum to the strictly ordered pairs $i < j$:

$$T_{\mu\mu}(\mathbf{r}, \mathbf{r}') = \frac{2V^2}{N^2} \sum_{i=0}^{N_\mu - 1} \sum_{j=i+1}^{N_\mu} R_{ij}(\mathbf{r}, \mathbf{r}'), \tag{20}$$

with R_{ij} the two–delta average derived in the previous section.

For a homogeneous block of species μ with Kuhn length b_μ, the connector mean–square between monomers i and j is

$$\sum_{k=i+1}^{j} b_k^2 = b_\mu^2\, (j - i). \tag{21}$$

Continuum limit for long chains

For a long chain ($N_\mu \to \infty$) we pass to contour variables. Let f_μ be the fraction of monomers belonging to species μ; then the double sum becomes a double integral,

$$\sum_{i=0}^{N_\mu - 1} \sum_{j=i+1}^{N_\mu} \longrightarrow \int_0^{f_\mu N} ds \int_s^{f_\mu N} ds'. \tag{22}$$

Inserting the real–space form of R_{ij} (a normalized Gaussian with variance proportional to $s' - s$) yields

$$T_{\mu\mu}(\mathbf{r} - \mathbf{r}') = \frac{2V^2}{N^2} \left(\frac{1}{V}\right) \int_0^{f_\mu N} ds \int_s^{f_\mu N} ds' \left(\frac{3}{2\pi b_\mu^2 (s' - s)}\right)^{3/2} \exp\left[-\frac{3\,|\mathbf{r} - \mathbf{r}'|^2}{2\, b_\mu^2 (s' - s)}\right]. \tag{23}$$

Fourier transform in the separation

Define the Fourier transform in the convention $\widehat{f}(\mathbf{q}) = \int d\mathbf{y}\, e^{-i\mathbf{q}\cdot\mathbf{y}} f(\mathbf{y})$, with $\mathbf{y} = \mathbf{r} - \mathbf{r}'$. Introduce the shorthand

$$\alpha = \frac{3}{2b_\mu^2 (s' - s)}, \qquad y = |\mathbf{y}|, \tag{24}$$

so that the Gaussian kernel is $(\alpha/\pi)^{3/2} e^{-\alpha y^2}$. Then

$$\widehat{T}_{\mu\mu}(\mathbf{q}) = \frac{2V}{N^2} \int_0^{f_\mu N} ds \int_s^{f_\mu N} ds' \left(\frac{\alpha}{\pi}\right)^{3/2} \int d\mathbf{y}\, e^{-\alpha y^2} e^{-i\mathbf{q}\cdot\mathbf{y}}. \tag{25}$$

The Gaussian integral over $\mathbf{y}$ is standard:

$$\int d\mathbf{y}\, e^{-\alpha y^2} e^{-i\mathbf{q}\cdot\mathbf{y}} = \left(\frac{\pi}{\alpha}\right)^{3/2} \exp\left[-\frac{|\mathbf{q}|^2}{4\alpha}\right]. \tag{26}$$

After cancellation of $(\alpha/\pi)^{3/2}$ with $(\pi/\alpha)^{3/2}$ we obtain

$$\widehat{T}_{\mu\mu}(\mathbf{q}) = \frac{2V}{N^2}\int_0^{f_\mu N} ds \int_s^{f_\mu N} ds' \, \exp\left[-\frac{b_\mu^2|\mathbf{q}|^2}{6}(s'-s)\right]. \tag{27}$$

Change variables and integrate

Let $u = s' - s$ (for fixed s), so $u \in [0, f_\mu N - s]$. Define

$$a := \frac{b_\mu^2|\mathbf{q}|^2}{6}. \tag{28}$$

The inner integral becomes

$$\int_0^{f_\mu N - s} du\, e^{-au} = \frac{1 - e^{-a(f_\mu N - s)}}{a}. \tag{29}$$

Therefore

$$\widehat{T}_{\mu\mu}(\mathbf{q}) = \frac{2V}{N^2 a}\int_0^{f_\mu N} ds \left[1 - e^{-a(f_\mu N - s)}\right]. \tag{30}$$

Carrying out the remaining s–integral,

$$\int_0^{f_\mu N} ds \left[1 - e^{-a(f_\mu N - s)}\right] = f_\mu N - \frac{1}{a}\Big(1 - e^{-a f_\mu N}\Big), \tag{31}$$

and thus

$$\widehat{T}_{\mu\mu}(\mathbf{q}) = \frac{2V}{N^2 a}\left[f_\mu N - \frac{1}{a}\Big(1 - e^{-a f_\mu N}\Big)\right]. \tag{32}$$

Final compact form

Introduce the dimensionless wavevector variable

$$x := aN = \frac{b_\mu^2|\mathbf{q}|^2 N}{6} = R_{g\mu}^2|\mathbf{q}|^2 \tag{33}$$

Then (32) becomes

$$\widehat{T}_{\mu\mu}(x) = \frac{2V}{x^2}\Big[f_\mu x + e^{-f_\mu x} - 1\Big]. \tag{34}$$

Different–species case and use of the derivation R_{ij}

Consider a diblock chain with an A-block of length $N_A = fN$ (Kuhn length b_A) followed by a B-block of length $N_B = (1-f)N$ (Kuhn length b_B), where $\epsilon = b_A/b_B$. We now evaluate the cross–species density correlation

$$T_{AB}(\mathbf{r}, \mathbf{r}') = \big\langle \rho_A(\mathbf{r})\, \rho_B(\mathbf{r}')\big\rangle_0. \tag{35}$$

In terms of monomer positions,

$$T_{AB}(\mathbf{r}, \mathbf{r}') = T_{\mu\nu}(\mathbf{r}, \mathbf{r}') = \frac{V^2}{N^2}\sum_{i=0}^{fN}\sum_{j=fN+1}^{N} R_{ij}(\mathbf{r}, \mathbf{r}'), \tag{36}$$

where i runs over A and j over B. The same two–delta average R_{ij} derived previously is used here, but now the sum of Kuhn lengths between i and j contains contributions from two blocks:

$$\sum_{k=i+1}^{j} b_k^2 = \underbrace{\sum_{k=i+1}^{fN} b_A^2}_{A\text{-segment}} + \underbrace{\sum_{k=fN+1}^{j} b_B^2}_{B\text{-segment}} = b_A^2\,(fN - i) + b_B^2\,(j - fN). \tag{37}$$

Continuum form in real space

Passing to contour variables $s \in [0, fN]$ and $s' \in [fN, N]$ (long-chain limit), and inserting the real-space form of R_{ij} gives

$$T_{AB}(\mathbf{r}-\mathbf{r}') = \frac{V^2}{N^2}\left(\frac{1}{V}\right)\int_0^{fN} ds \int_{fN}^{N} ds' \left[\frac{3}{2\pi\big(b_A^2(fN-s)+b_B^2(s'-fN)\big)}\right]^{3/2} \times \exp\left[-\frac{3\,|\mathbf{r}-\mathbf{r}'|^2}{2\big(b_A^2(fN-s)+b_B^2(s'-fN)\big)}\right]. \tag{38}$$

We Fourier transform with respect to the separation $\mathbf{y} = \mathbf{r} - \mathbf{r}'$:

$$\widehat{T}_{AB}(\mathbf{q}) := \int d\mathbf{y}\, e^{-i\mathbf{q}\cdot\mathbf{y}}\, T_{AB}(\mathbf{y}). \tag{39}$$

Insert (38):

$$\widehat{T}_{AB}(\mathbf{q}) = \frac{V}{N^2}\int_0^{fN} ds \int_{fN}^{N} ds' \left[\frac{3}{2\pi\, S(s,s')}\right]^{3/2} \int d\mathbf{y}\ \exp\left[-\frac{3\,y^2}{2S(s,s')} - i\mathbf{q}\cdot\mathbf{y}\right], \tag{40}$$

where for brevity we set

$$S(s,s') := b_A^2\,(fN-s) + b_B^2\,(s'-fN). \tag{41}$$

Introduce

$$\alpha(s,s') := \frac{3}{2\,S(s,s')}, \qquad \Longrightarrow \qquad \left[\frac{3}{2\pi\, S}\right]^{3/2} = \left(\frac{\alpha}{\pi}\right)^{3/2}, \quad \frac{1}{4\alpha} = \frac{S}{6}. \tag{42}$$

The inner Gaussian integral over $\mathbf{y}$ is then

$$\int d\mathbf{y}\ \exp\big[-\alpha y^2 - i\mathbf{q}\cdot\mathbf{y}\big] = \left(\frac{\pi}{\alpha}\right)^{3/2}\exp\left[-\frac{q^2}{4\alpha}\right] = \left(\frac{\pi}{\alpha}\right)^{3/2}\exp\left[-\frac{q^2 S(s,s')}{6}\right]. \tag{43}$$

The prefactors $(\alpha/\pi)^{3/2}$ and $(\pi/\alpha)^{3/2}$ cancel exactly, leaving

$$\widehat{T}_{AB}(\mathbf{q}) = \frac{V}{N^2}\int_0^{fN} ds \int_{fN}^{N} ds'\ \exp\left[-\frac{q^2}{6}\, S(s,s')\right]. \tag{44}$$

Now use the linearity of $S(s,s')$ in s and s', and define $a_A = \frac{b_A^2|\mathbf{q}|^2}{6}$ and $a_B = \frac{b_B^2|\mathbf{q}|^2}{6}$.:

$$\frac{q^2}{6}\, S(s,s') = \underbrace{\frac{b_A^2 q^2}{6}}_{=:a_A}(fN-s) \ + \ \underbrace{\frac{b_B^2 q^2}{6}}_{=:a_B}(s'-fN), \tag{45}$$

so the exponential factorizes:

$$e^{-\frac{q^2}{6}S(s,s')} = e^{-a_A(fN-s)}\ e^{-a_B(s'-fN)}. \tag{46}$$

Because the integrand is now a product of a function of s and a function of s', the double integral separates into the product of two one–dimensional integrals:

$$\widehat{T}_{AB}(\mathbf{q}) = \frac{V}{N^2}\left[\int_0^{fN} ds\ e^{-a_A(fN-s)}\right]\left[\int_{fN}^{N} ds'\ e^{-a_B(s'-fN)}\right]. \tag{47}$$

Evaluating these elementary exponentials gives

$$\int_0^{fN} ds\ e^{-a_A(fN-s)} = \frac{1-e^{-a_A fN}}{a_A}, \qquad \int_{fN}^{N} ds'\ e^{-a_B(s'-fN)} = \frac{1-e^{-a_B(1-f)N}}{a_B}, \tag{48}$$

and substituting (48) into (47) yields

$$\widehat{T}_{AB}(\mathbf{q}) = \frac{V}{N^2}\,\frac{1-e^{-a_A fN}}{a_A}\,\frac{1-e^{-a_B(1-f)N}}{a_B}, \tag{49}$$

Dimensionless form and compact result

Introduce the dimensionless variables

$$x := a_B N = \frac{b_B^2 |\mathbf{q}|^2 N}{6}, \qquad \alpha := \frac{b_A^2}{b_B^2} = \epsilon^2 \quad \Rightarrow \quad a_A N = \alpha x. \tag{50}$$

Then $a_A f N = \alpha f x$ and $a_B(1-f)N = (1-f)x$, so

$$\widehat{T}_{AB}(\mathbf{q}) = \frac{V}{\alpha\, x^2}\left(1 - e^{-\alpha f x}\right)\left(1 - e^{-(1-f)x}\right), \qquad x = \frac{b_B^2 |\mathbf{q}|^2 N}{6}, \quad \alpha = \frac{b_A^2}{b_B^2}. \tag{51}$$

Checks and comments.

- **Symmetry under block exchange.** Exchanging $A \leftrightarrow B$ amounts to $f \mapsto 1-f$ and $\alpha \mapsto \alpha^{-1}$ together with redefining x with b_A; (51) is invariant under this relabeling.
- **Equal Kuhn lengths.** For $b_A = b_B$ ($\alpha = 1$) the result simplifies to $\widehat{T}_{AB} = \frac{V}{x^2}\big(1 - e^{-fx}\big)\big(1 - e^{-(1-f)x}\big)$, the familiar cross–term of the Debye form for a symmetric Gaussian diblock.